

# Quantitative intra-Golgi transport and organization data suggest the stable compartment nature of the Golgi

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## Abstract

How the intra-Golgi secretory transport works remains a mystery. The cisternal progression and the stable compartment models have been proposed and are under debate. Classic cisternal progression model posits that both the intra-Golgi transport and Golgi exit of secretory cargos should occur at a constant velocity dictated by the cisternal progression; furthermore, COPI-mediated intra-Golgi retrograde transport is essential for maintaining the Golgi organization. Leveraging our recently developed Golgi imaging tools in nocodazole-induced Golgi ministacks, we found that the intra-Golgi transport velocity of a secretory cargo decreases during their transition from the cis to the trans-side of the Golgi, and different cargos exhibit distinct velocities even within the same cisternae. We observed a vast variation in the Golgi residence times of different cargos. Remarkably, truncation of the luminal domain causes the Golgi residence time of Tac — a standard transmembrane secretory cargo without intra-Golgi recycling signals — to extend from 16 minutes to a notable 3.4 hours. Additionally, when COPI-mediated intra-Golgi retrograde transport was inhibited by brefeldin A, we found that nocodazole-induced Golgi can remain stacked for over 30 - 60 minutes. Therefore, our findings challenge the classical cisternal progression model and suggest the stable compartment nature of the Golgi.

### eLife assessment

This **important** study sheds new light on cargo movement within the Golgi apparatus, challenging the cisternal progression model by providing **convincing** evidence for a velocity decrease from cis to trans Golgi and variable speeds within cisternae, suggesting a more stable compartmental nature. While these findings propose refinements to the classic model, they prompt further exploration of recent models like rapid partitioning and rim progression, necessitating additional experimental approaches to account for cargo expression variations and HeLa cell-specific effects.

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## Introduction

The Golgi complex in mammalian cells plays a crucial role in membrane trafficking and post-translational modification of proteins and lipids. Centrally positioned around the microtubule organization center (Glick and Luini, 2011; Klumperman, 2011), it comprises laterally connected Golgi stacks, each containing 4 – 7 tightly adjacent membrane sacs known as cisternae. Conventionally, a Golgi stack is recognized to have three regions: the *cis*, medial, and *trans*-Golgi. The *trans*-Golgi network (TGN) assembles outside the *trans*-side of a Golgi stack and is mainly composed of tubular and vesicular membranes (De Matteis and Luini, 2008). In the secretory pathway, newly synthesized proteins and lipids (cargos) exit the ER at the ER exit site (ERES), pass the ER and Golgi intermediate compartment (ERGIC), and reach the Golgi. Cargos subsequently transit from the *cis*-side through the medial and eventually reach the *trans*-side of the Golgi, where they exit the Golgi and target the plasma membrane (Tie et al., 2016).

Despite decades of research, the Golgi remains one of the most enigmatic organelles, particularly regarding the mechanism of intra-Golgi transport of secretory cargos (Emr et al., 2009; Glick and Luini, 2011). Two primary intra-Golgi transport models have been proposed and are currently the subject of debate. In the classic cisternal progression or maturation model, secretory cargos passively reside within Golgi cisternae, which progress or mature from the *cis* to the *trans*-Golgi cisternae to facilitate the forward or anterograde transport of cargos. Simultaneously, post-translational modification enzymes such as glycosyltransferases move in reverse or retrograde transport, altering cisternal properties to become the next cisternae of the Golgi stack. Under this perspective, the Golgi functions at a dynamic equilibrium of anterograde and retrograde intra-Golgi transport. The model predicts that all secretory cargos should have the same and constant intra-Golgi transport and Golgi exit velocity. It can provide a plausible explanation for the intra-Golgi transition of oversized secretory cargos (Bonfanti et al., 1998; Mironov et al., 2001). Moreover, direct observations of cisternal maturation have been made in budding yeast *Saccharomyces cerevisiae* (Kurokawa et al., 2019; Losev et al., 2006; Matsuura-Tokita et al., 2006), although similar observation has not been reported in mammalian cells. However, the budding yeast Golgi differs significantly from the mammalian one in the cisternal organization – it scatters throughout the cytoplasm as unstacked compartments. This substantial difference casts doubt on the general applicability of the budding yeast Golgi observation to higher eukaryotes. In contrast, the stable compartment model posits that Golgi cisternae are stable entities. During intra-Golgi transport, carriers actively move secretory cargos from one cisterna or compartment to the next, from the *cis* to the *trans*-side, while post-translational modification enzymes remain stationary. At the *trans*-side of the Golgi, cargos are sorted into carriers bound for the plasma membrane. As a result, different cargos can exhibit distinct intra-Golgi transport and Golgi exit velocities under this model.

The ongoing debate between these two models underscores the complexity of the Golgi and reflects the insufficiency of the existing experimental data, especially the intra-Golgi transport kinetics data. Two semi-quantitative approaches exist for studying the intra-Golgi transport kinetics of ER-synchronized secretory cargos. The first is the biochemical approach, where intra-Golgi transport kinetics can be indirectly deduced by subtracting ER-to-Golgi and Golgi-to-plasma membrane transport from the overall secretion (ER-to-plasma membrane transport) during the chase. However, this indirect approach provides only an averaged intra-Golgi transport kinetics, making it incapable of measuring the instantaneous intra-Golgi transport velocity at a sub-Golgi region. The second approach involves electron microscopy (EM) imaging of immuno-gold labeled secretory cargos at Golgi cisternae (Bezoussenko et al., 2014; Trucco et al., 2004). The distribution of gold particles among individual Golgi cisternae during the chase directly indicates

intra-Golgi transport kinetics. Nevertheless, this approach demands specialized techniques and equipment and considerable manual work in EM imaging and subsequent image examination, limiting its widespread use in other labs.

The limitations in current approaches call for novel methods, especially ones based on fluorescence microscopy, to resolve the intra-Golgi transport spatially and kinetically. However, optically resolving the intra-Golgi secretory transport in mammalian cells is challenging due to the thinness (200 – 400 nm) and random orientation of Golgi stacks. To overcome this, we leverage the uniform and rotationally symmetrical arrangement of nocodazole-induced Golgi ministacks (hereafter referred to as Golgi ministacks). Extensive studies have provided strong evidence that ministacks obtained under prolonged nocodazole treatment ( $\geq 3$  hours), the condition employed in our studies, largely represent the native Golgi stack (Cole et al., 1996 [DOI](#); Fourriere et al., 2016 [DOI](#); Rogalski et al., 1984 [DOI](#); Tie et al., 2018 [DOI](#); Trucco et al., 2004 [DOI](#); Van De Moortele et al., 1993 [DOI](#)). We developed a numerical Golgi localization tool called Golgi localization by imaging centers of mass (GLIM) that can precisely pinpoint a Golgi protein's cisternal localization with nanometer accuracy in Golgi ministacks (Tie et al., 2017 [DOI](#); Tie et al., 2016 [DOI](#)). Alongside this, we propose using the metric of Golgi residence time to quantify a Golgi protein's retention in the Golgi (Sun et al., 2021 [DOI](#); Sun et al., 2020 [DOI](#)).

Here, we utilized past and newly acquired GLIM and Golgi residence time data to analyze intra-Golgi transport and Golgi exit kinetics quantitatively. Our data revealed that neither intra-Golgi transport nor Golgi exit exhibits a constant velocity. We discovered that when the luminal domain of Tac — a conventional transmembrane secretory cargo lacking intra-Golgi recycling signals — is truncated, its Golgi residence time increases from 16 min to a substantial 3.4 h. Through GLIM, we examined the cisternal organization of Golgi ministacks under brefeldin A (BFA) treatment, which halts retrograde intra-Golgi transport. Remarkably, under this condition, we found that nocodazole-induced Golgi ministacks can remain stacked for 30 - 60 min. Therefore, our findings underscore the stable nature of Golgi cisternae. They challenge the classical cisternal progression model and favor the stable compartment model.

## Results and discussion

### The intra-Golgi transport is not a motion with constant velocity

The classic cisternal progression model postulates that the intra-Golgi transport velocity of a secretory cargo should remain constant. However, this prediction has not been directly tested due to the inherent challenge of measuring the intra-Golgi transport velocity. The advent of GLIM has allowed us to address this issue. In GLIM, briefly, HeLa cells expressing the RUSH secretory cargo (RUSH reporter) (Boncompain et al., 2012 [DOI](#)) and GalT-mCherry, a *trans*-Golgi marker containing amino acids 1-81 of B4GALT1, were initially treated with nocodazole and cycloheximide (a protein synthesis inhibitor). Subsequently, cells were chased in biotin with nocodazole and cycloheximide for various lengths of time ( $t$ ) before immuno-fluorescence labeling for endogenous GM130, a *cis*-Golgi marker. Except for TNF $\alpha$ -SBP-GFP, which uses Ii-Streptavidin as the ER hook, all RUSH reporters employ signal sequence fused streptavidin-KDEL as their ER hook. Additionally, apart from SBP-GFP, a soluble secretory protein, all RUSH reporters we selected are transmembrane proteins. We calculated centers of mass as the positions of GM130, RUSH cargo, and GalT-mCherry within each analyzable Golgi ministack. The RUSH cargo's Golgi localization quotient, or  $LQ$ , is calculated by dividing its distance from GM130 by GalT-mCherry's distance from GM130. The  $LQ$  is a linear numerical metric to indicate a cargo's axial localization within the Golgi, with a nanometer range of precision (Tie et al., 2017 [DOI](#); Tie et al., 2016 [DOI](#)). We previously linearly defined regions of Golgi: ERES/ERGIC ( $LQ < -0.25$ ), *cis* ( $-0.25 \leq LQ < 0.25$ ), medial ( $0.25 \leq LQ < 0.75$ ), *trans*-Golgi ( $0.75 \leq LQ < 1.25$ ), and TGN ( $LQ \geq 1.25$ ).

To analyze the intra-Golgi transport kinetics of secretory cargos, we measured the  $LQ$ s of RUSH reporters after various durations of biotin administration (chase). Most kinetic data was previously reported (Tie et al., 2017 [DOI](#); Tie et al., 2016 [DOI](#)) and re-analyzed here. Additionally, we generated new data and replicated specific measurements. **Figures 1** [DOI](#), **Supplementary Figure 1** [DOI](#), and **Supplementary File 1** [DOI](#) illustrate that our  $LQ$  vs. time plots are highly reproducible. As previously reported, all  $LQ$  vs. time plots fit the following first-order exponential function (Equation 1 [DOI](#)) well with an adjusted  $R^2$  (adj.  $R^2$ )  $\geq 0.91$  (Fig. 1A-J [DOI](#) and Supplementary Fig. 1A-M [DOI](#), left panels).

$$LQ = y_0 - Ae^{(-\ln 2 \frac{t}{t_{\text{intra}}})} \quad (\text{Equation 1})$$

In Equation 1 [DOI](#),  $t$  represents chase time in minutes (biotin treatment starts at  $t = 0$ );  $\ln 2$  is the natural logarithm of 2;  $A$  is a constant;  $y_0$  represents the  $LQ$  of the Golgi exit site;  $t_{\text{intra}}$  is the time that the cargo reaches half of the transport range and is hereafter referred to as the intra-Golgi transport time. We define the instantaneous intra-Golgi transport velocity as the derivative of  $LQ$  with respect to time,  $dLQ/dt$ , which measures the axial transport velocity of the center of the mass of the synchronized cargo wave. It should also follow the first-order exponential function (Equation 2 [DOI](#)).

$$\frac{dLQ}{dt} = \ln 2 \frac{A}{t_{\text{intra}}} e^{(-\ln 2 \frac{t}{t_{\text{intra}}})} \quad (\text{Equation 2})$$

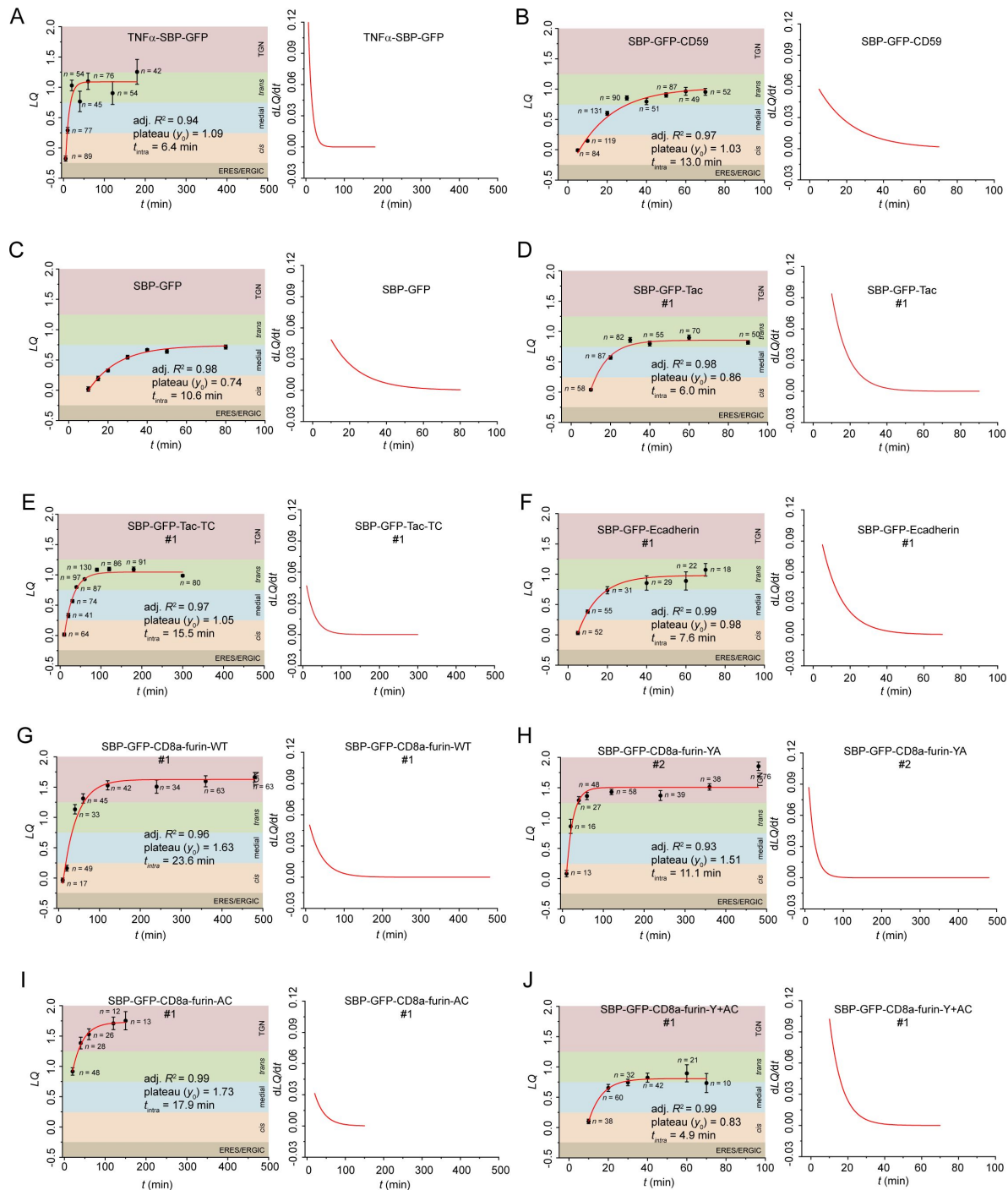
In this context, the intra-Golgi transport velocity is the highest when the cargo enters the secretory pathway ( $t = 0$ ). However, it is crucial to approach this extrapolation cautiously due to the lack of experimental data at  $t \leq 5$  min, when a RUSH reporter's high ER background and low Golgi signal make it challenging to select analyzable Golgi ministacks. It is evident that the  $dLQ/dt$  of all our RUSH reporters slows to zero as they transit across the Golgi stack to reach  $LQ = y_0$  at the *trans*-Golgi (Figure. 1A-J [DOI](#), right panels), where we propose RUSH reporters exit Golgi ministacks in carriers en route to the plasma membrane (Tie et al., 2018 [DOI](#); Tie et al., 2016 [DOI](#); Tie et al., 2022 [DOI](#)). Hence, the intra-Golgi transport velocity of a secretory cargo does not remain constant, contradicting the prediction of the classic cisternal progression model.

## Distinct intra-Golgi transport velocities for different cargos at the same cisternae

From Equations 1 [DOI](#) and 2 [DOI](#), we derive the following relationship (Equation 3 [DOI](#)):

$$\frac{dLQ}{dt} = \frac{\ln 2}{t_{\text{intra}}} (y_0 - LQ) \quad (\text{Equation 3})$$

In our previous work, through side-averaging, we determined that one  $LQ$  unit corresponds to 274 nm (Tie et al., 2022 [DOI](#)). Hence, we scaled Equation 3 [DOI](#) by 274 nm to derive the instantaneous intra-Golgi transport velocity in nm/min. When plotting the instantaneous intra-Golgi transport velocity (nm/min) against the  $LQ$  for selected RUSH reporters, as seen in Figure 2A [DOI](#), we observed that different secretory cargos exhibit varied transport velocities even within the same cisternae or at the same  $LQ$  values. At  $LQ = 0.4$ , corresponding to the medial-Golgi region, our RUSH reporters' instantaneous intra-Golgi transport velocities are calculated in Table 1 [DOI](#) according to Equation 3 [DOI](#). For instance, instantaneous intra-Golgi transport velocities of SBP-GFP, SBP-GFP-Tac-TC, SBP-GFP-CD59, SBP-GFP-Tac, and TNF $\alpha$ -SBP-GFP are 6.1, 8.0, 9.2, 14.6, and 20.5 nm/min, respectively (Table 1 [DOI](#)). These findings highlighted that different secretory cargos possess distinct intra-Golgi transport velocities within the same Golgi cisternae, challenging the prediction made by the classic cisternal progression model.



**Figure 1**

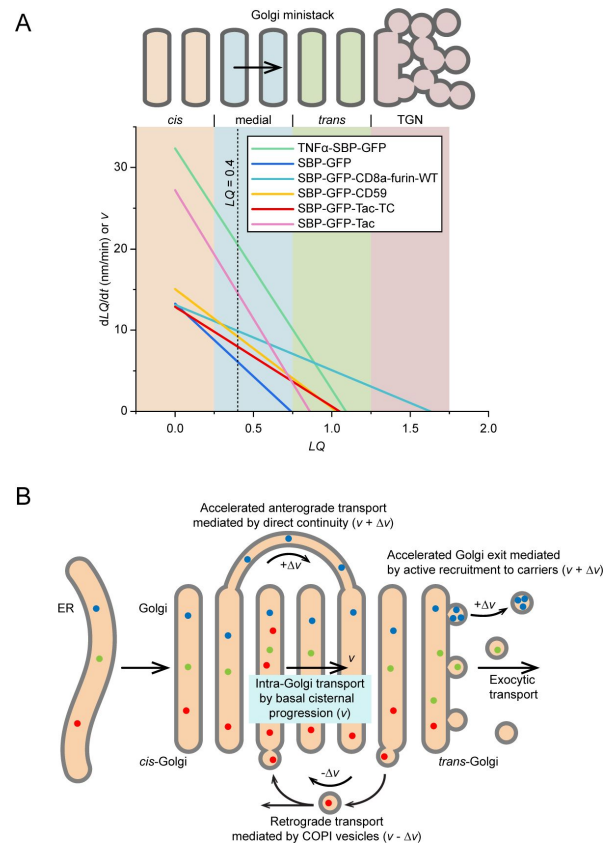
The intra-Golgi transport kinetics for RUSH Reporters in Golgi ministacks. HeLa cells transiently co-expressing individual RUSH reporter and GalT-mCherry were incubated with 33  $\mu\text{M}$  nocodazole for 3 hours. This was followed by a chase with 33  $\mu\text{M}$  nocodazole, 10  $\mu\text{g}/\text{ml}$  cycloheximide, and 40  $\mu\text{M}$  biotin for variable time durations ( $t$ ) before fixation and GM130 immunostaining. Except for TNF $\alpha$ -SBP-GFP, which uses Ii-Streptavidin as the ER hook, all RUSH reporters employ signal sequence fused streptavidin-KDEL as their ER hook. Using GLIM,  $LQ$ s for the RUSH reporters were then calculated and plotted. Each panel has an  $LQ$  vs. time plot on the left, with  $n$  (the number of analyzable Golgi ministacks), adj.  $R^2$ ,  $y_0$ , and  $t_{\text{intra}}$  indicated. On the right side of each panel, the  $dLQ/dt$  vs. time plot represents the first-order derivative of its corresponding  $LQ$  vs. time plot on the left. ERES/ERGIC ( $LQ < -0.25$ ), cis ( $-0.25 \leq LQ < 0.25$ ), medial ( $0.25 \leq LQ < 0.75$ ), trans-Golgi ( $0.75 \leq LQ < 1.25$ ), and TGN ( $1.25 \leq LQ$ ) zones were color-shaded. Aside from the one in panel C, all  $LQ$  vs. time plots are from our previous reports (see [Table 1](#)).

| RUSH reporter                           | $t_{\text{intra}}$<br>(min) | $y_0$ | Adj.<br>$R^2$ | dLQ/dt (at<br>$LQ = 0.4$ ) | dLQ/dt (at<br>$LQ = 0.4$ )<br>(nm/min) |
|-----------------------------------------|-----------------------------|-------|---------------|----------------------------|----------------------------------------|
| TNF $\alpha$ -SBP-GFP <sup>1</sup>      | 6.4                         | 1.09  | 0.94          | 0.075                      | 20.5                                   |
| SBP-GFP-CD59 <sup>1</sup>               | 13.0                        | 1.03  | 0.97          | 0.034                      | 9.2                                    |
| SBP-GFP                                 | 10.6                        | 0.74  | 0.98          | 0.022                      | 6.1                                    |
| SBP-GFP-Tac #1 <sup>2</sup>             | 6.0                         | 0.86  | 0.98          | 0.053                      | 14.6                                   |
| SBP-GFP-Tac #2                          | 8.2                         | 0.91  | 0.98          | 0.043                      | 11.8                                   |
| SBP-GFP-Tac-TC #1 <sup>2</sup>          | 15.5                        | 1.05  | 0.97          | 0.029                      | 8.0                                    |
| SBP-GFP-Tac-TC #2                       | 19.1                        | 1.16  | 0.94          | 0.028                      | 7.6                                    |
| TfR-SBP-GFP                             | 7.0                         | 1.16  | 0.99          | 0.075                      | 20.6                                   |
| SBP-GFP-E-cadherin #1 <sup>1</sup>      | 7.6                         | 0.98  | 0.99          | 0.053                      | 14.5                                   |
| SBP-GFP-E-cadherin #2 <sup>1</sup>      | 8.7                         | 1.15  | 0.96          | 0.060                      | 16.5                                   |
| SBP-GFP-E-cadherin #3 <sup>1</sup>      | 6.5                         | 1.11  | 0.98          | 0.075                      | 20.6                                   |
| SBP-GFP-CD8a-furin-WT #1 <sup>1</sup>   | 23.6                        | 1.63  | 0.96          | 0.036                      | 9.9                                    |
| SBP-GFP-CD8a-furin-WT #2 <sup>1</sup>   | 18.7                        | 1.58  | 0.97          | 0.044                      | 12.0                                   |
| SBP-GFP-CD8a-furin-WT #3 <sup>1</sup>   | 13.1                        | 1.55  | 0.97          | 0.061                      | 16.7                                   |
| SBP-GFP-CD8a-furin-YA #1 <sup>1</sup>   | 5.9                         | 1.49  | 0.98          | 0.128                      | 35.0                                   |
| SBP-GFP-CD8a-furin-YA #2 <sup>1</sup>   | 11.1                        | 1.51  | 0.93          | 0.069                      | 19.0                                   |
| SBP-GFP-CD8a-furin-YA #3 <sup>1</sup>   | 16.3                        | 1.48  | 0.95          | 0.046                      | 12.6                                   |
| SBP-GFP-CD8a-furin-AC #1 <sup>1</sup>   | 17.9                        | 1.73  | 0.99          | 0.052                      | 14.1                                   |
| SBP-GFP-CD8a-furin-AC #2 <sup>1</sup>   | 26.9                        | 1.50  | 1.00          | 0.028                      | 7.8                                    |
| SBP-GFP-CD8a-furin-AC #3 <sup>1</sup>   | 23.2                        | 1.58  | 0.96          | 0.035                      | 9.7                                    |
| SBP-GFP-CD8a-furin-Y+AC #1 <sup>1</sup> | 4.9                         | 0.83  | 0.99          | 0.061                      | 16.7                                   |
| SBP-GFP-CD8a-furin-Y+AC #2 <sup>1</sup> | 7.1                         | 1.00  | 0.99          | 0.059                      | 16.0                                   |
| SBP-GFP-CD8a-furin-Y+AC #3 <sup>1</sup> | 7.7                         | 1.04  | 0.92          | 0.058                      | 15.8                                   |

**Table 1**

Intra-Golgi transport kinetics of secretory RUSH reporters in Golgi ministacks. See the legend of **Figure 1** for details. #1-3 indicate independent replicates. Except for TNF $\alpha$ -SBP-GFP, which uses Ii-Streptavidin as the ER hook, all RUSH reporters employ signal sequence fused streptavidin-KDEL as their ER hook. Superscripts 1 and 2 indicate data were re-analyzed from previous publications (Tie et al., 2016) and (Sun et al., 2020). WT, wild type.  $t_{\text{intra}}$  (intra-Golgi transport time),  $y_0$ , and adj.  $R^2$  were calculated by fitting measured LQ vs. time kinetics data to Equation 1. dLQ/dt (at LQ = 0.4) was calculated by Equation 3, and converted to nm/min by multiplying 274 nm per LQ unit. For furin cytosolic tail RUSH reporters, YA is YKGL, a tyrosine-based motif, to AKGL mutation, while AC is SDSEDE, an acidic cluster sequence, to ADAAAAA mutation (Tie et al., 2016). YA + AC indicates mutations in both sites (Tie et al., 2016).





**Figure 2**

Different secretory cargos exhibit distinct intra-Golgi transport velocities within the same cisternae of Golgi ministacks. (A) Plots of  $dLQ/dt$  vs.  $LQ$  for selected RUSH reporters reveal diverse intra-Golgi transport velocities at the same cisternae. These plots were constructed based on [Equation 3](#), using parameters from [Table 1](#). Different regions within the Golgi — *cis* ( $-0.25 \leq LQ < 0.25$ ), medial ( $0.25 \leq LQ < 0.75$ ), *trans*-Golgi ( $0.75 \leq LQ < 1.25$ ), and TGN ( $1.25 \leq LQ$ ) — are color-shaded for easier identification. To convert  $dLQ/dt$  to nm/min, we multiplied  $dLQ/dt$  by 274 nm per  $LQ$  unit. A dotted vertical line marks the  $LQ$  value of 0.40. (B) Modifying the classic cisternal progression model to explain the distinct intra-Golgi transport velocities at the same cisternae. According to this model, Golgi cisternae progress at a constant velocity ( $v$ ). Direct-continuity-based anterograde transport could accelerate the intra-Golgi transport velocity ( $v + \Delta v$ ), while the COPI-mediated retrograde transport could reduce it ( $v - \Delta v$ ).

## Golgi residence times vary significantly among different secretory cargos

Once transmembrane secretory cargos transit the Golgi stack, they reach the *trans*-side of the Golgi and are packed into carriers destined for the plasma membrane. The classic cisternal progression model posits that secretory cargos linearly depart the *trans*-Golgi. Contrary to this, cargo exit kinetics have been shown to adhere to a first-order exponential relationship rather than a linear one (Hirschberg et al., 1998 [DOI](#); Patterson et al., 2008 [DOI](#); Sun et al., 2020 [DOI](#)). In an attempt to reconcile these observations, Patterson *et al.* introduced the rapid partitioning model, which stands distinct from the classic cisternal progression and stable compartment models (Patterson et al., 2008 [DOI](#)). This model suggests that cargos rapidly diffuse throughout the Golgi, subsequently segregating into multiple post-translational processing and export domains, where cargos are packed into carriers bound for the plasma membrane. Nonetheless, synchronized traffic waves have been observed through various techniques, including EM (Trucco et al., 2004 [DOI](#)) and advanced light microscopy methods we developed, such as GLIM and side-averaging (Tie et al., 2016 [DOI](#); Tie et al., 2022 [DOI](#)). These findings suggest that the rapid partitioning model might not accurately represent the true nature of the intra-Golgi transport.

In both the cisternal progression and stable compartment models, a rapid transit through the Golgi stack, combined with a rate-limiting step at the Golgi exit, could result in the observed exponential Golgi exit kinetics (Luini, 2011 [DOI](#)). While we could hypothetically apply such a rate-limiting step to both models to rationalize the exponential kinetics of cargo exit, the classic cisternal progression model encounters more significant challenges. Firstly, the first-order exponential Golgi exit implies that clearing a synchronized wave of secretory cargo from the *trans*-cisternae would take an indefinite time, which is inconsistent with the transient nature of the *trans*-cisternae as described by the classic cisternal progression model. Secondly, since cargos are considered passive in the classic cisternal progression model, the exit kinetics, as measured by the Golgi residence times, should be the same across all secretory cargos.

The Golgi residence time is a cargo's duration at the *trans*-Golgi cisternae before exit and is a metric for Golgi retention (Sun et al., 2021 [DOI](#); Sun et al., 2020 [DOI](#)). The first step to determine this metric involves synchronizing a transmembrane protein at the Golgi. For Golgi transmembrane resident proteins like glycosyltransferases and secretory transmembrane proteins with a substantial Golgi localization at the steady state, such as GFP-Tac-TC, synchronization is not required. However, for secretory transmembrane cargos lacking a significant Golgi pool at the steady state, such as TfR-GFP, GFP-Tac, and RUSH reporters like TNF $\alpha$ -SBP-GFP, synchronization is achieved through a 20°C temperature block. Subsequently, live imaging at 37 °C is performed to capture the Golgi fluorescence intensity decay of the protein in the presence of cycloheximide. The Golgi residence time is calculated as the half-time ( $t_{1/2}$ ) by fitting the intensity decay to a first-order exponential function.

Our extensive measurement demonstrated that Golgi residence times of secretory cargos display a wide range of values (Table 2 [DOI](#)). For example, TNF $\alpha$ -SBP-GFP has a Golgi residence time of  $6.0 \pm 0.4$  min ( $n = 73$ ), one of the shortest, while SBP-GFP-CD59 has a Golgi residence time of  $16 \pm 2$  min ( $n = 29$ ) (Supplementary Fig. 2A [DOI](#)). Hence, our data suggest that different secretory cargos reside at the *trans*-Golgi cisternae for varying durations, contradicting predictions from the classic cisternal progression model.



| Transmembrane protein              | Golgi residence time ( $t_{1/2}$ ) | <i>n</i> | SEM     |
|------------------------------------|------------------------------------|----------|---------|
| TNF $\alpha$ -SBP-GFP <sup>1</sup> | 6.0 min                            | 73       | 0.4 min |
| GFP-CD8a <sup>1</sup>              | 7.8 min                            | 26       | 0.7 min |
| TfR-GFP <sup>1</sup>               | 10 min                             | 12       | 1 min   |
| GFP-Tac <sup>1</sup>               | 16 min                             | 45       | 2 min   |
| SBP-GFP-CD59                       | 16 min                             | 29       | 2 min   |
| GFP-Tac (5A) <sup>1</sup>          | 47 min                             | 57       | 3 min   |
| GFP-CD8a-TC                        | 66 min                             | 19       | 7 min   |
| GFP-Tac-TC <sup>1</sup>            | 3.4 h                              | 22       | 0.3 h   |
| MGAT2-GFP <sup>2</sup>             | 4.9 h                              | 26       | 0.5 h   |
| ST6GAL1-GFP <sup>1</sup>           | 5.3 h                              | 21       | 0.6 h   |

**Table 2**

Golgi residence times of transmembrane secretory cargos and Golgi glycosyltransferases in native Golgi (without nocodazole). Superscripts 1 and 2 indicate data were from previous publications (Sun et al., 2020 [DOI](#)) and (Sun et al., 2021 [DOI](#)), respectively. The RUSH reporter, TNF $\alpha$ -SBP-GFP, employs signal sequence fused streptavidin-KDEL as the ER hook. *n*, the number of quantified cells; SEM, standard error of the mean.

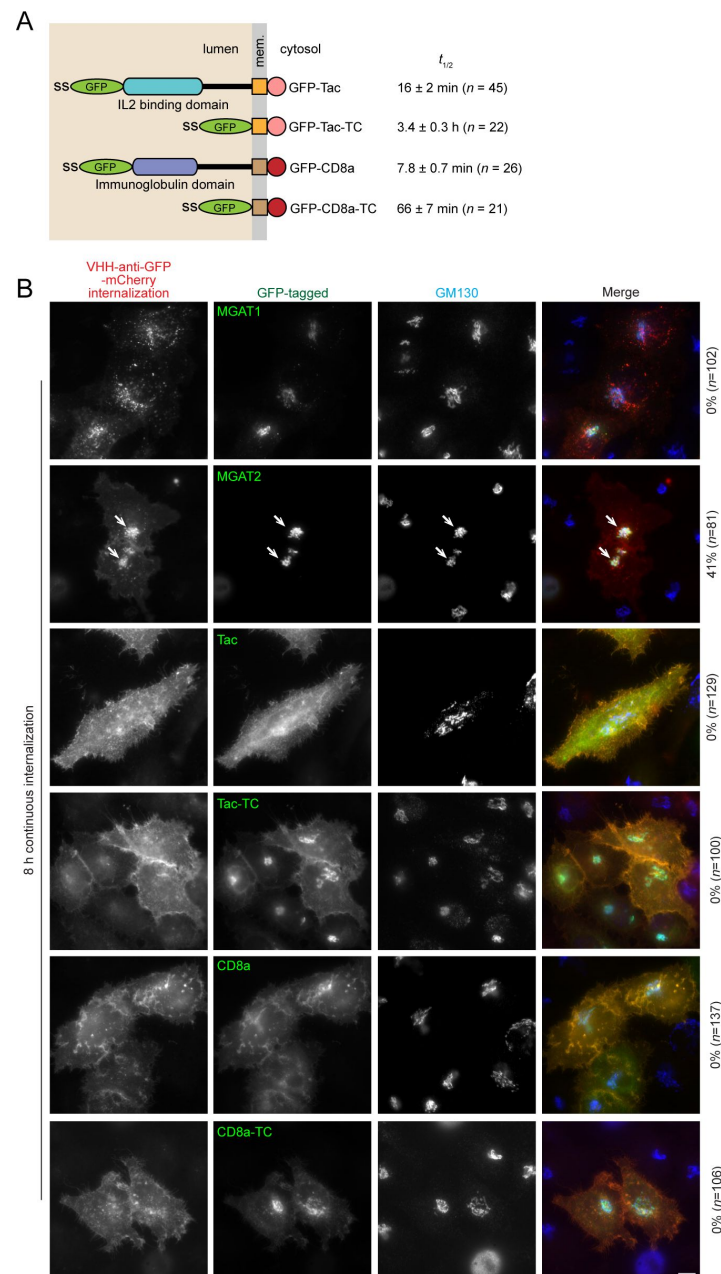
## Cargos exhibiting prolonged Golgi residence times suggest the *trans*-Golgi interior might be a stable domain

The classic cisternal progression model could be modified to explain the diverse transport kinetics. For example, accelerated anterograde or retrograde transport might be introduced on top of the basal cisternal progression to account for the wide range of intra-Golgi transport velocities and Golgi residence times we observed (**Fig. 2B**). Accelerated anterograde transport mechanisms might include continuity-based direct diffusion across cisternae via heterologous cisternal connections (Bezoussenko et al., 2014; Marsh et al., 2004; Trucco et al., 2004). Hence, secretory cargo with such a mechanism would have a faster intra-Golgi transport velocity. This mechanism might explain the rapid and diverse intra-Golgi transport velocity of secretory cargos such as VSVG, insulin, albumin, and alpha1-antitrypsin (Bezoussenko et al., 2014; Marsh et al., 2004; Trucco et al., 2004). Similarly, active recruitment to exocytic carriers budding at the *trans*-Golgi possibly might shorten the Golgi residence time. On the other hand, COPI-coated carriers might facilitate retrograde intra-Golgi transport to counter the cisternal progression, accounting for the slow intra-Golgi transport velocity and prolonged Golgi residence time of cargos. Indeed, COPI has been known to maintain certain glycosyltransferases' Golgi retention by direct or indirect interactions (Ali et al., 2012; Eckert et al., 2014; Liu et al., 2018; Pereira et al., 2014; Rizzo et al., 2021; Schmitz et al., 2008; Tu et al., 2008). However, such a retrograde mechanism requires the interaction between secretory cargo and the COPI coat.

Earlier, we identified truncation mutants of two typical secretory cargos, GFP-Tac and GFP-CD8a, which reside in the Golgi nearly as stably as Golgi glycosyltransferases (Sun et al., 2020). These are type I transmembrane proteins comprising from their N- to C-termini a signal peptide, GFP, luminal domain, transmembrane domain (TMD), and cytosolic tail (**Fig. 3A**). Previously, we and others demonstrated that fully glycosylated luminal domain can function as a Golgi exit signal (Gut et al., 1998; Sun et al., 2020). Their Golgi residence time was documented at  $16 \pm 2$  min ( $n = 45$ ) and  $7.8 \pm 0.7$  min ( $n = 26$ ), respectively (**Table 2**). However, after truncating the luminal domain, the Golgi residence time for the resultant chimeras, GFP-Tac-TC and GFP-CD8a-TC, extended significantly to  $3.4 \pm 0.3$  h ( $n = 22$ ) and  $66 \pm 7$  min ( $n = 19$ ) (**Table 2**) (**Supplementary Fig. 2B**), respectively. Consequentially, their long Golgi residence times ensure a significant steady-state Golgi pool.

The following evidence suggests that GFP-Tac-TC behaves like a bona fide Golgi resident transmembrane protein. Firstly, its Golgi residence time, 3.4 h, is comparable to that of a typical Golgi glycosyltransferase, such as ST6GAL1 ( $5.3 \pm 0.6$  h,  $n = 21$ ) (Sun et al., 2021). Second, GFP-Tac-TC follows GalT-mCherry to localize to the ER under the BFA treatment reversibly (Sun et al., 2020). Third, GFP-Tac-TC's  $LQ$ ,  $0.94 \pm 0.02$  ( $n = 118$ ), combined with its cisternal interior localization (Sun et al., 2020), indicates that it primarily resides within the interior of the *trans*-cisternae, similar to many Golgi glycosyltransferases (Tie et al., 2018).

According to the modified cisternal progression model above, GFP-Tac-Tc could have an active retrieval mechanism mediated by an unidentified signal in its cytosolic tail or TMD to counter the cisternal progression. Considering that GFP-Tac-TC's  $LQ$ ,  $0.94 \pm 0.02$  ( $n = 118$ ), is close to its Golgi exit site,  $y_0$  ( $0.94 - 1.05$ ) (**Table 1**; **Fig. 1E** and **Supplementary Fig. 1C**), measured by its RUSH version, there are two possible retrieval pathways, acting at either post-Golgi or intra-Golgi stages. To test if GFP-Tac-Tc possesses a Golgi retrieval pathway after its Golgi exit, we incubated HeLa cells transiently expressing MGAT1-GFP (negative control) (Sun et al., 2021), MGAT2-GFP (positive control) (Sun et al., 2021), GFP-Tac, or GFP-Tac-TC with a recombinant mCherry-fused anti-GFP nanobody (VHH-anti-GFP-mCherry) continuously for 8 h (**Fig. 3B**). We found that VHH-anti-GFP-mCherry localized to the Golgi in a significant fraction of MAGT2-GFP-expressing cells (41%,  $n = 81$ ) but not in MGAT1-GFP cells (0%,  $n = 102$ ). This result is consistent with our previous



**Figure 3**

GFP-Tac-TC and GFP-CD8a-TC might not have a Golgi retrieval mechanism once exiting the Golgi. (A) Domain organization of GFP-tagged Tac, Tac-TC, CD8a, and CD8a-TC. ss, signal sequence. Mem., membrane.  $t_{1/2}$  values are from [Table 2](#) and shown as mean  $\pm$  SEM.  $n$ , the number of cells analyzed. (B) The plasma membrane-localized GFP-Tac-TC and GFP-CD8a-TC are not retrieved to the Golgi. HeLa cells transiently expressing indicated GFP-tagged protein were incubated continuously with VHH-anti-GFP-mCherry for 8 h before fixation and immunostaining for the endogenous Golgi marker, GM130. Images were acquired under a wide-field microscope. The internalized VHH-anti-GFP-mCherry has negative Golgi localization for all GFP-tagged proteins except for MGAT2 (arrows), a positive control. The percentage of cells showing the Golgi localization of VHH-anti-GFP-mCherry is labeled at right.  $n$ , the number of cells analyzed. Scale bar, 10  $\mu$ m.

report that MGAT2, but not MAGT1, has a Golgi retrieval mechanism (Sun et al., 2021), although its molecular mechanism is still unknown. In contrast, VHH-anti-GFP-mCherry did not localize to the Golgi in cells expressing other constructs, including GFP-Tac-TC, suggesting that GFP-Tac-TC might not possess a post-Golgi retrieval mechanism targeting the Golgi. Next, we reason that given GFP-Tac's short Golgi residence time ( $16 \pm 2$  min,  $n = 45$ ), its cytosolic tail and TMD might not facilitate any active intra-Golgi retrograde transport mechanism, such as COPI coat binding, to recycle GFP-Tac-TC from Golgi exiting. Since GFP-Tac and GFP-Tac-TC share identical TMD and cytosolic tail sequences, if such a retrograde mechanism existed, GFP-Tac would have a similar Golgi residence time to GFP-Tac-TC. The same observation and reasoning also apply to GFP-CD8a-TC (Fig. 3B). Therefore, we argue that GFP-Tac-TC and GFP-CD8a-TC might not have retrieval signal to facilitate their Golgi residence, although proving a protein does not possess a transport signal is challenging. Since the cisternal interior is continuous with the cisternal rim both in membrane and lumen, our findings suggest that the cisternal interior at the *trans*-Golgi might be a stable domain. Our findings suggest that retention within the *trans*-side stable domain, rather than continuous retrieval to counter the cisternal progression (treadmilling), could be the primary mechanism for the long Golgi residence times of GFP-Tac-TC and GFP-CD8a-TC.

We have previously demonstrated in fluorescence microscopy that during their intra-Golgi transition, small cargos, such as CD59, E-cadherin, and VSVG, localize to the cisternal interior, coinciding with Golgi glycosyltransferases, while large cargos, such as FM4 aggregates and collagenX, position themselves at the cisternal rim, where trafficking machinery components localize (Tie et al., 2018). The rim partitioning of large secretory cargos is consistent with previous EM studies of procollagen I (Bonfanti et al., 1998), FM4 aggregates (Lavie et al., 2013), and algal protein aggregates (Engel et al., 2015). Therefore, these findings might support a modified version of the stable compartment model — the rim progression model (Lavie et al., 2013; Pfeffer, 2010; Volchuk et al., 2000). In this model, each Golgi cisterna seems to have a stable interior domain and a dynamic rim domain; the rim domain of Golgi cisternae undergoes constant en bloc fission and fusion for intra-Golgi cargo transport. The rim progression model can readily explain the diverse intra-Golgi transport velocities, the exponential Golgi exit, and the wide range of Golgi residence times of secretory cargos. It also helps explain the retention of Golgi glycosyltransferases at the cisternal interior since many do not recycle via COPI-coated vesicles (Liu et al., 2018) or possess a post-Golgi retrieval pathway (Sun et al., 2021). The Golgi glycosyltransferase at the cisternal interior might assemble a dense protein matrix based on fluorescence (Tie et al., 2018) and EM data (Engel et al., 2015). It is tempting to speculate that the glycosyltransferase matrix could functionally mirror a gel-filtration chromatography matrix with a defined porosity. Hence, large secretory cargos, such as FM4 aggregates and collagenX, are excluded from the interior, where the glycosyltransferase matrix localizes, while small secretory cargos can enter and become kinetically trapped there.

## The Golgi maintains its stacked organization after 30 min BFA treatment

COPI functions in the retrograde direction at the ER-Golgi interface and within the Golgi (Glick and Luini, 2011; Popoff et al., 2011; Rabouille and Klumperman, 2005). According to the classic cisternal progression model, COPI-mediated retrograde intra-Golgi transport recycles resident transmembrane proteins, such as glycosyltransferases and transport machinery components. This model predicts that upon the compromise of COPI, intra-Golgi recycling would stop, and the Golgi stack would continuously lose its materials and eventually disappear, depending on the cisternal progression rate. In addition to its retrograde role, COPI has also been documented to function in the anterograde ER-to-Golgi transport (Monetta et al., 2007; Weigel et al., 2021).

To test the role of COPI in the Golgi organization, we employed BFA, a small-molecule fungal metabolite that rapidly dissociates COPI and clathrin from the Golgi, inhibits the ER-to-Golgi trafficking, and causes the fusion of the Golgi with the ER in 10 - 20 min (Klausner et al., 1992). It

does so by rapidly inactivating class I ARFs, which recruit COPI and clathrin coats to the Golgi membrane (Donaldson et al., 1992 [↗](#); Helms and Rothman, 1992 [↗](#)). In contrast to the native Golgi, nocodazole-induced Golgi ministacks have been documented to be more resistant to BFA, and, hence, the disappearance of Golgi occurs at a much later time (> 30 min) (Lippincott-Schwartz et al., 1990 [↗](#)). However, it is unclear if the nocodazole-induced Golgi maintains a similar stacked organization under the BFA treatment.

Given this, we studied the *LQs* of several Golgi markers under 30 – 60 min BFA treatment (Fig. 4 [↗](#)). The fast dissociation of Arf1-GFP from Golgi ministacks confirmed the effectiveness of BFA (Supplementary Fig. 3 [↗](#)). We also observed that extended BFA treatment considerably reduced intact Golgi ministacks. We measured the *LQs* of five transmembrane Golgi markers, GFP-golgin-84, GS15, ST6GAL1-GFP, TGN46, CD8a-furin, and CD8a-CI-M6PR (Fig. 4 [↗](#)).

GS15's *LQ* was monitored for 60 min, with the rest for 30 min. We observed that *LQs* of ST6GAL1 and GS15 did not change significantly, while the *LQ* of golgin-84 increased from 0.2 to 0.4. Remarkably, *LQs* of all three TGN markers decreased and approached that of the *trans*-Golgi, suggesting a collapse of the TGN, possibly due to the dissociation of clathrin and its adaptor proteins. In summary, despite dramatic changes in the TGN, *LQs* of tested Golgi stack markers still follow a similar arrangement to those of pre-BFA treatment. Therefore, our data demonstrate that the Golgi maintains its stacked organization for at least 30 – 60 min, even in the absence of COPI-mediated intra-Golgi retrograde transport. Our findings suggest that the Golgi might not be a dynamic equilibrium between the cisternal progression and retrograde trafficking and argue for a by-default stable nature of the Golgi stack. It is worth noting that the cellular effect of BFA is complex and pleiotropic. For example, in addition to Arfs, it can inhibit lipid metabolic enzymes (De Matteis et al., 1994 [↗](#)). So, the disassociation of COPI might not be the sole factor responsible for our observations.

## Summary

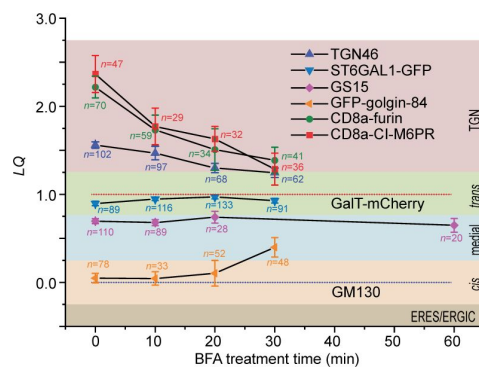
We introduced new quantitative data on the intra-Golgi transport dynamics. However, our study has limitations. First, our approach relied on the overexpression of fluorescence protein-tagged cargos. The synchronized release of a large amount of cargo could significantly saturate and skew the intra-Golgi transport. Second, we utilized nocodazole-induced ministacks instead of the native Golgi to analyze the intra-Golgi transport, which could raise concerns about the impact of depolymerizing microtubules on the intra-Golgi transport and Golgi organization.

Our findings suggest that the Golgi cisternal interior might be a stable domain, therefore supporting a modified version of the stable compartment model, the rim progression model. However, we do not think our data alone can resolve the two models, which have been the subject of debate for several decades. Further refinement of the classic cisternal progression model might also account for our data. We anticipate further development using our approach will provide more systematic data to test and refine future intra-Golgi transport models. Moreover, we hope that our study will stimulate further research into this longstanding and intriguing question.

## Materials and methods

### DNA plasmids, antibodies, and small molecules

To clone RUSH reporter, SBP-GFP, two PCRs were performed using li-Strep<sub>ss</sub>-SBP-EGFP-Ecadherin (a gift plasmid from F. Perez) (Boncompain et al., 2012 [↗](#)) and pEGFP-C1 (Clontech) as templates and the following primer pairs (GAT GCA CCC GGG AGG CGC GCC ATG and CTC CTC GCC CTT GCT CAC ACC TGC AGG TGG TTC ACG) and (CGT GAA CCA CCT GCA GGT GTG AGC AAG GGC GAG GAG and GAT GCA TCT AGA TTA CTT GTA CAG CTC GTC CAT), respectively. The mixture of the two purified PCR fragments was used as the template for the third round of PCR amplification using



**Figure 4**

The organization of Golgi ministacks under the BFA treatment. HeLa cells transiently co-expressing GalT-mCherry and indicated GFP or CD8a-tagged Golgi proteins were incubated with 33  $\mu$ M nocodazole for 3 hours. This was followed by additional treatment with 33  $\mu$ M nocodazole and 1  $\mu$ M BFA for variable durations before fixation. Endogenous GS15 and TGN46 were immunostained. LQs of different Golgi proteins were then determined using GLIM. The dotted lines represent LQs for GM130 ( $LQ = 0.00$ ) and GalT-mCherry ( $LQ = 1.00$ ). The total number of analyzed Golgi ministacks is denoted by 'n'.



the first and the fourth primer listed above. The resulting PCR fragment was digested by XmaI and XbaI and ligated to XmaI and XbaI digested li-Strep<sub>ss</sub>-SBP-EGFP-Ecadherin DNA plasmid. To clone GFP-CD8a-TC, the coding sequence of the TMD and cytosolic tail of CD8a was amplified by PCR using GFP-CD8a(Sun et al., 2020 [DOI](#)) as a template and the following primers, CAG TGC CTC GAG GAC TTC GCC TGT GAT ATC TA and GAC CGT GAA TTC TTA GAC GTA TCT CGC CGA AAG GCT G. The resulting PCR fragment was digested by EcoRI and XhoI and ligated to EcoRI and XhoI digested GFP-CD8a DNA plasmid.

ST6GAL1-GFP (ST-GFP) (Sun et al., 2020 [DOI](#)), CD8a-furin(Mahajan et al., 2013 [DOI](#)), CD8a-CI-M6PR(Mahajan et al., 2013 [DOI](#)) were previously described. RUSH Tfr-SBP-GFP was a gift plasmid from J. Bonifacino (Chen et al., 2017 [DOI](#)). GFP-golgin-84 (Diao et al., 2003 [DOI](#)) was a gift plasmid from M. Lowe. Arf1-GFP(Wessels et al., 2006 [DOI](#)) was a gift plasmid from F. van Kuppeveld.

Mouse monoclonal antibody anti-CD8a (OKT8) was from the hybridoma culture supernatant. Mouse monoclonal antibodies against GM130 (#610822) and GS15 (#610960) were purchased from BD Bioscience. Rabbit polyclonal antibody against TGN46 was from Abcam (#ab50595). Alexa Fluor 488, 594, and 647-conjugated goat anti-mouse or anti-rabbit secondary antibodies were purchased from Thermo Fisher Scientific.

## Small molecules

Nocodazole (#487928; working concentration: 33  $\mu$ M), brefeldin A (working concentration: 10  $\mu$ g/ml), and cycloheximide (working concentration: 10  $\mu$ g/ml) were purchased from Merck, Epicenter Technologies, and Sigma Aldrich, respectively.

## Cell culture and transfection

HeLa cells were maintained in Dulbecco's Modified Eagle's Medium (DMEM) supplemented with 10% fetal bovine serum (FBS) to make it complete DMEM. For cell transfection, Lipofectamine 2000 (Thermo Fisher Scientific) was used per the manufacturer's instructions. For BFA treatment, cells were treated with complete DMEM containing 1  $\mu$ M BFA for the specified duration. For nocodazole treatment, cells were treated with complete DMEM containing 33  $\mu$ M nocodazole for 3 hours to induce the formation of Golgi ministacks.

## Immunofluorescence

Cells for immunofluorescence were grown on No. 1.5  $\Phi$ 12 mm glass coverslips. They were fixed using 4% paraformaldehyde in phosphate-buffered saline (PBS). Following a PBS wash to remove residual paraformaldehyde, any remaining paraformaldehyde within the cells was neutralized with 100 mM NH<sub>4</sub>Cl. The cells were then processed for immunofluorescence labeling by first incubating with mouse or rabbit primary antibodies, followed by Alexa Fluor 488, 594, and/or 647 conjugated goat anti-mouse or anti-rabbit secondary antibodies. Both primary and secondary antibodies were diluted in PBS containing 5% fetal bovine serum, 2% bovine serum albumin, and 0.1% saponin (Sigma-Aldrich). The labeled cells were mounted in the Mowiol mounting medium, composed of 12% Mowiol 4-88 (EMD Millipore), 30% glycerol, and 100 mM Tris pH 8.5. After the mounting medium had dried, the coverslips were sealed with nail polish and stored at - 20°C.

## Acquiring LQs

To analyze LQs of intra-Golgi transport RUSH reporters, HeLa cells transiently co-expressing individual GFP-tagged RUSH reporter and GalT-mCherry were cultured in complete DMEM supplemented with 16 nM His-tagged streptavidin (in-house purified using Addgene #20860, a gift plasmid from A. Ting)(Howarth et al., 2006 [DOI](#)). Following a 3 h nocodazole treatment, cells were chased with 40  $\mu$ M biotin, 10  $\mu$ g/ml cycloheximide, and 33  $\mu$ M nocodazole for various durations before fixation.

In another set of experiments to study *LQs* of Golgi markers under the BFA treatment, nocodazole-treated HeLa cells transiently expressing GalT-mCherry were further incubated with 1  $\mu\text{M}$  BFA and 33  $\mu\text{M}$  nocodazole for various durations before fixation. For these experiments, the Golgi markers were either co-expressed with GalT-mCherry as a GFP-tagged construct or detected as an endogenous protein by immunostaining.

The methodology for acquiring *LQs* through GLIM has been described in our previous studies (Tie et al., 2017 [DOI](#); Tie et al., 2016 [DOI](#)). Briefly, cells were further immuno-labeled to visualize endogenous GM130. Ministacks exhibiting fluorescence signals of GM130, transfected GalT-mCherry, and the testing protein were imaged using a wide-field microscope. Ministacks were manually selected for analysis, and fluorescence centers for GM130, GalT-mCherry, and the testing protein were acquired using Fiji (<https://imagej.net/software/fiji/> [DOI](#)). After chromatic aberration correction, coordinates of fluorescence centers were used to calculate *LQs*, defined as the ratio of axial distances from GM130 to the testing protein and from GM130 to GalT-mCherry. Intra-Golgi transport kinetic *LQ* data were fitted to the first-order exponential function using OriginPro 2020.

## Acquisition of Golgi residence times

The methodology follows protocols previously described (Sun et al., 2021 [DOI](#); Sun et al., 2020 [DOI](#)). Nocodazole was not used in these experiments. Briefly, HeLa cells on a  $\Phi$  35mm glass-bottom Petri dish were transiently transfected to co-express a GFP-tagged reporter and GalT-mCherry. For RUSH reporters, cells were treated with 40  $\mu\text{M}$  biotin at 20 °C for 2 hours to accumulate the reporter at the Golgi. Live imaging was performed in a  $\text{CO}_2$ -independent medium (Thermo Fisher Scientific) with 10% FBS, 4 mM glutamine, and 10  $\mu\text{g/ml}$  cycloheximide, using a wide-field microscope until the cellular GFP fluorescence at the Golgi nearly vanished. The resulting time-lapse images were segmented based on GalT-mCherry using Fiji. Total GFP fluorescence within the Golgi was quantified and fitted to the first-order exponential function  $y = y_0 + A1\exp(-x/t_1)$  in OriginPro 2020. Golgi residence time,  $t_{1/2}$ , was calculated as  $0.693 \times t_1$ . We only included time-lapse data with adj.  $R^2 \geq 0.80$  and acquisition length  $\geq 1.33 \times t_{1/2}$ .

## VHH-anti-GFP-mCherry internalization assay

6  $\times$  His-tagged VHH-anti-GFP-mCherry was purified as previously described (Sun et al., 2021 [DOI](#); Sun et al., 2020 [DOI](#)). In the internalization assay, HeLa cells transiently expressing GFP-tagged reporters were continuously incubated with 5  $\mu\text{g/ml}$  VHH-anti-GFP-mCherry at 37 °C for 8 h. After washing, cells were fixed and imaged.

## Wide-field microscopy

*LQs* and Golgi residence times were measured using a wide-field microscope based on Olympus IX83. The microscope featured a 100 $\times$  oil objective lens (NA 1.40), a motorized stage for sample positioning, and automated filter cubes to accommodate different fluorescence channels. Dichroic mirrors and filters were optimized for GFP/Alexa fluor 488, mCherry/Alexa Fluor 594, and Alexa Fluor 647. Imaging was captured with an sCMOS (scientific complementary metal oxide semiconductor) camera (Neo) by Andor. A 200W metal halide light source (Lumen Pro 200) by Prior Scientific provided illumination. Operational control and data collection were facilitated through Metamorph software by Molecular Devices.

## Airyscan microscopy

The Airyscan microscopy was performed using a Zeiss LSM710 confocal microscope, equipped with alpha Plan-Apochromat 100 $\times$  NA 1.46 objective and the Airyscan module (Carl Zeiss). The system operation was controlled by Zen software (Carl Zeiss). Two lines of laser lights were used, 488 and 640 nm. The emission band was selected to optimize the capture of the emission light while minimizing channel crosstalk.

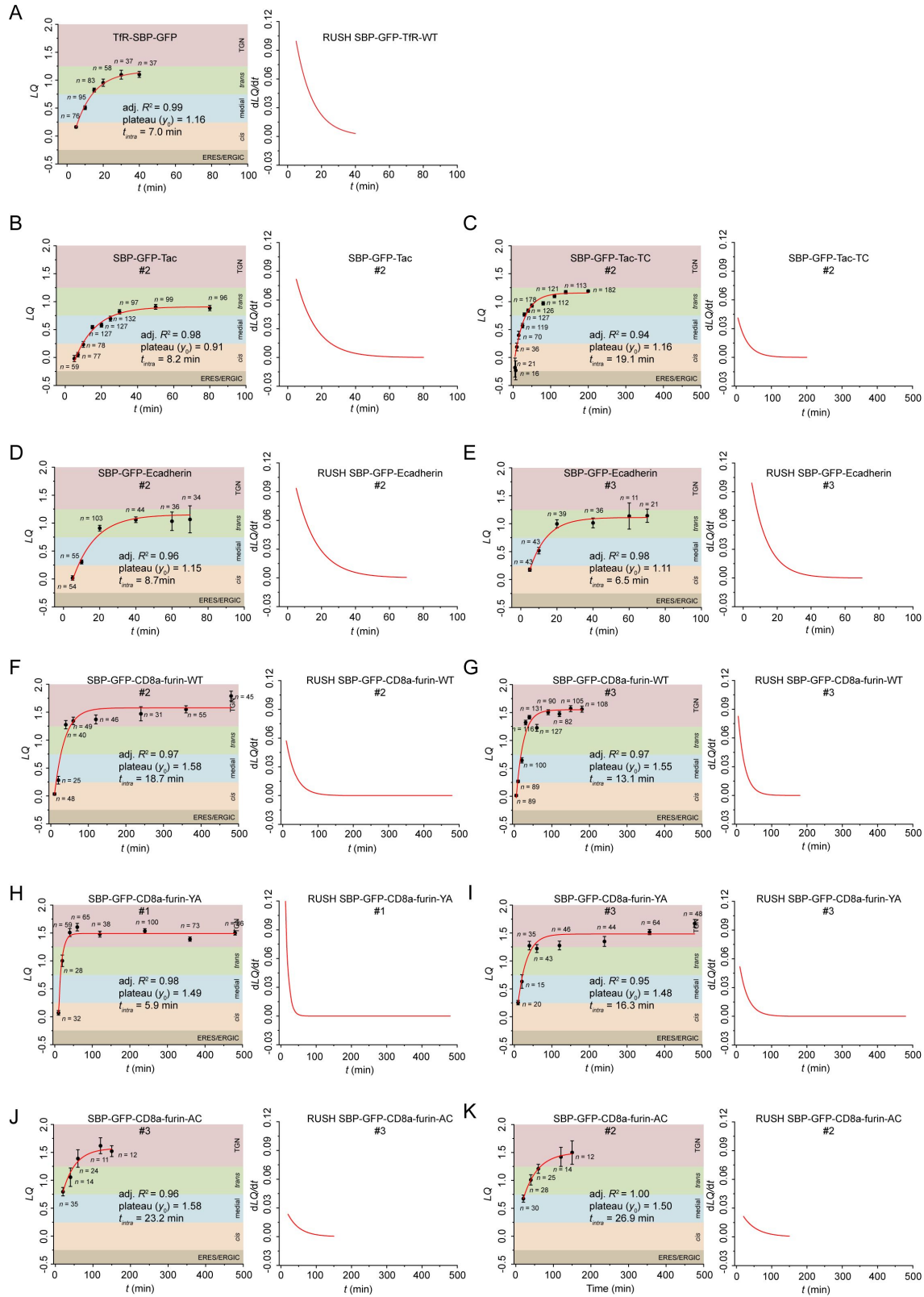
## Declaration of interests

The authors declare no competing interests.

## Acknowledgements

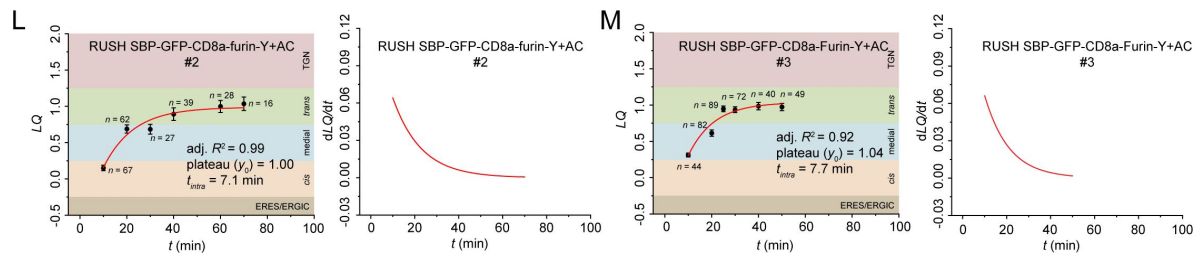
We want to thank J. Bonifacino (National Institute of Health, USA), F. van Kuppeveld (Utrecht University), M. Lowe (University of Manchester, UK), F. Perez (Institute of Curie, France), and A. Ting (Stanford University, USA) for sharing DNA plasmids. This project is supported by the Ministry of Education, Singapore, under its Tier 2 MOE-T2EP30221-0001 and Tier 1 RG 25/22.

## Supplementary information

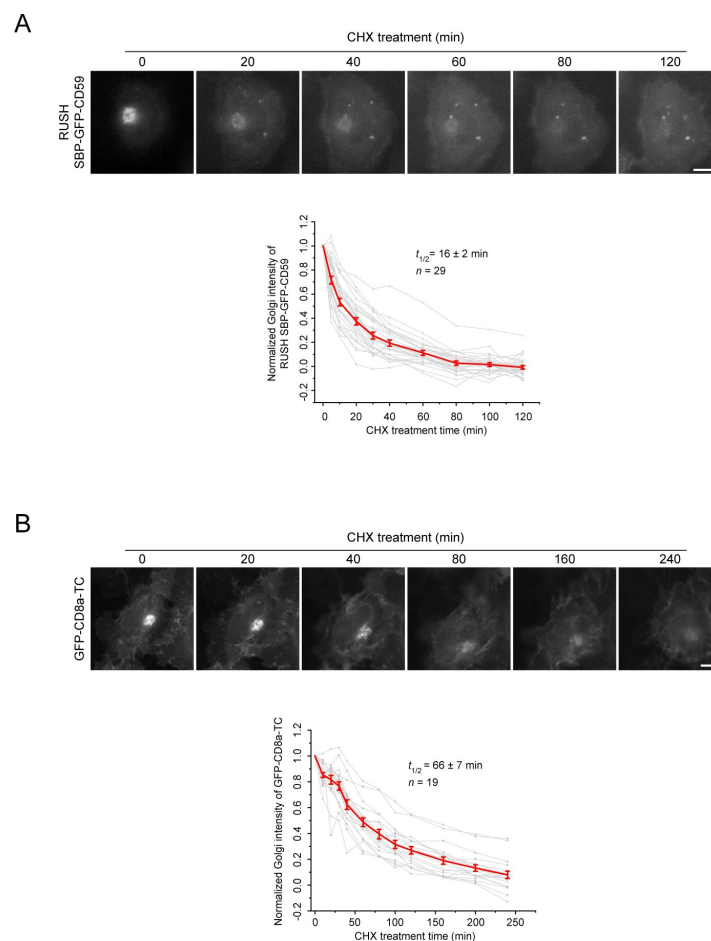


## Supplementary Figure 1

Additional intra-Golgi transport kinetics of RUSH reporters. All LQ vs. time plots, except those in panels (A-C), are from our previous reports (see [Table 1](#)). See [Figure 1](#)'s legend for details.

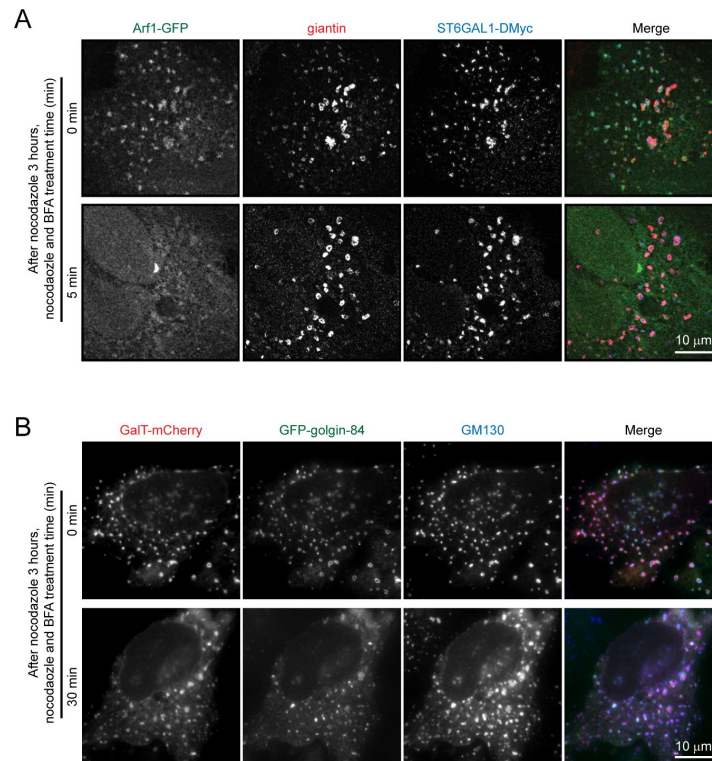


**Supplementary Figure 1** (continued)



**Supplementary Figure 2**

Acquiring Golgi residence times of RUSH SBP-GFP-CD59 and GFP-CD8a-TC. (A) HeLa cells were transiently transfected to co-express RUSH SBP-GFP-CD59 and GalT-mCherry. Cells were incubated with 40  $\mu$ M biotin and 10  $\mu$ g/ml cycloheximide at 20  $^{\circ}$ C to accumulate the RUSH reporter in the Golgi before live-cell imaging under a wide-field microscope at 37  $^{\circ}$ C. Only RUSH SBP-GFP-CD59 images are shown. (B) HeLa cells transiently co-expressing GFP-CD8a-TC and GalT-mCherry were treated with cycloheximide and imaged live under a wide-field microscope. Only GFP-CD8a-TC images are shown. In both (A) and (B), the total fluorescence intensity within the Golgi was quantified, normalized, and plotted against the time. Each intensity series was fitted to the first-order exponential function to calculate  $t_{1/2}$ . Grey and red indicate individual and averaged time series, respectively. CHX, cycloheximide. Error bar, mean  $\pm$  stand error of the mean.  $n$ , the number of quantified cells. Scale bar, 10  $\mu$ m.



### Supplementary Figure 3

The localization of Arf1 and golgin-84 under the treatment of nocodazole and BFA. (A) HeLa cells transiently expressing Arf1-GFP and ST6GAL1-DMyc were treated with 33 µM nocodazole for 3 hours. This was followed by additional treatment with 33 µM nocodazole and 1 µM BFA for 5 min before fixation and immunostaining of Myc-tag and giantin. Images were acquired by Airyscan microscopy. (B) HeLa cells transiently expressing GalT-mCherry and GFP-golgin-84 were treated with 33 µM nocodazole for 3 hours. This was followed by additional treatment with 33 µM nocodazole and 1 µM BFA for 30 min before fixation and immunostaining for GM130. Images were acquired by wide-field microscopy. Scale bar, 10 µm.



Fig. 1

A. SBP-TNF $\alpha$ -GFP

| Chase time (min) | <i>n</i> | <i>LQ</i> | <i>SEM</i> |
|------------------|----------|-----------|------------|
| 5                | 89       | -0.18     | 0.04       |
| 10               | 77       | 0.29      | 0.05       |
| 20               | 54       | 1.03      | 0.09       |
| 40               | 45       | 0.77      | 0.17       |
| 60               | 76       | 1.10      | 0.13       |
| 120              | 54       | 0.90      | 0.19       |
| 180              | 42       | 1.25      | 0.21       |

B. SBP-GFP-CD59

|    |     |       |      |
|----|-----|-------|------|
| 5  | 84  | -0.01 | 0.02 |
| 10 | 119 | 0.15  | 0.02 |
| 20 | 131 | 0.60  | 0.03 |
| 30 | 90  | 0.85  | 0.04 |
| 40 | 51  | 0.80  | 0.05 |
| 50 | 87  | 0.90  | 0.04 |
| 60 | 49  | 0.96  | 0.07 |
| 70 | 52  | 0.95  | 0.06 |

C. SBP-GFP

|    |     |      |      |
|----|-----|------|------|
| 10 | 67  | 0.02 | 0.04 |
| 15 | 84  | 0.19 | 0.04 |
| 20 | 124 | 0.33 | 0.02 |
| 30 | 138 | 0.55 | 0.03 |
| 40 | 207 | 0.67 | 0.02 |
| 50 | 105 | 0.64 | 0.03 |
| 80 | 82  | 0.71 | 0.03 |

D. SBP-GFP-Tac #1

|    |    |      |      |
|----|----|------|------|
| 10 | 58 | 0.04 | 0.02 |
| 20 | 87 | 0.57 | 0.03 |
| 30 | 82 | 0.86 | 0.04 |
| 40 | 55 | 0.80 | 0.04 |
| 60 | 70 | 0.90 | 0.04 |
| 90 | 50 | 0.82 | 0.03 |

E. SBP-GFP-Tac-TC #1

|     |     |      |      |
|-----|-----|------|------|
| 10  | 64  | 0.02 | 0.03 |
| 20  | 41  | 0.33 | 0.04 |
| 30  | 74  | 0.57 | 0.03 |
| 40  | 87  | 0.80 | 0.02 |
| 60  | 97  | 0.93 | 0.02 |
| 90  | 130 | 1.08 | 0.03 |
| 120 | 86  | 1.10 | 0.03 |
| 180 | 91  | 1.09 | 0.04 |
| 300 | 80  | 0.99 | 0.02 |

F. SBP-GFP-Ecadherin #1

|    |    |      |      |
|----|----|------|------|
| 5  | 52 | 0.03 | 0.03 |
| 10 | 55 | 0.38 | 0.03 |
| 20 | 31 | 0.74 | 0.06 |
| 40 | 29 | 0.85 | 0.12 |
| 60 | 22 | 0.89 | 0.15 |
| 70 | 18 | 1.07 | 0.10 |

|    |    |       |      |
|----|----|-------|------|
| 10 | 17 | -0.03 | 0.04 |
| 20 | 49 | 0.16  | 0.05 |

## Supplementary File 1

*LQ* data employed in plots presented in [Figure 1](#), [Figure 4](#), and [Supplementary Figure 1](#). *n*, the number of quantified cells. SEM, standard error of the mean.

|                             |     |    |      |      |
|-----------------------------|-----|----|------|------|
| G. SBP-GFP-CD8a-furin-WT #1 | 40  | 33 | 1.13 | 0.08 |
|                             | 60  | 45 | 1.31 | 0.08 |
|                             | 120 | 42 | 1.53 | 0.07 |
|                             | 240 | 34 | 1.51 | 0.11 |
|                             | 360 | 63 | 1.60 | 0.09 |
|                             | 480 | 63 | 1.66 | 0.08 |

|                             |     |    |      |      |
|-----------------------------|-----|----|------|------|
| H. SBP-GFP-CD8a-furin-YA #2 | 10  | 13 | 0.08 | 0.05 |
|                             | 20  | 16 | 0.86 | 0.12 |
|                             | 40  | 27 | 1.29 | 0.06 |
|                             | 60  | 48 | 1.36 | 0.06 |
|                             | 120 | 58 | 1.43 | 0.04 |
|                             | 240 | 39 | 1.37 | 0.08 |
|                             | 360 | 38 | 1.51 | 0.05 |
|                             | 480 | 76 | 1.85 | 0.07 |

|                             |     |    |      |      |
|-----------------------------|-----|----|------|------|
| I. SBP-GFP-CD8a-furin-AC #1 | 20  | 48 | 0.92 | 0.06 |
|                             | 40  | 28 | 1.38 | 0.10 |
|                             | 60  | 26 | 1.53 | 0.09 |
|                             | 120 | 12 | 1.71 | 0.10 |
|                             | 150 | 13 | 1.75 | 0.15 |

|                               |    |    |      |      |
|-------------------------------|----|----|------|------|
| J. SBP-GFP-CD8a-furin-Y+AC #1 | 10 | 38 | 0.10 | 0.04 |
|                               | 20 | 60 | 0.65 | 0.06 |
|                               | 30 | 32 | 0.74 | 0.05 |
|                               | 40 | 42 | 0.82 | 0.07 |
|                               | 60 | 21 | 0.89 | 0.14 |
|                               | 70 | 10 | 0.73 | 0.16 |

Fig. 4

| TGN46 | BFA treatment time (min) | <i>n</i> | <i>LQ</i> | <i>SEM</i> |
|-------|--------------------------|----------|-----------|------------|
|       | 0                        | 102      | 1.56      | 0.04       |
|       | 10                       | 97       | 1.47      | 0.08       |
|       | 20                       | 68       | 1.30      | 0.05       |
|       | 30                       | 62       | 1.25      | 0.05       |

|                |    |     |      |      |
|----------------|----|-----|------|------|
| ST6GAL1-moxGFP | 0  | 89  | 0.90 | 0.03 |
|                | 10 | 116 | 0.95 | 0.02 |
|                | 20 | 133 | 0.97 | 0.02 |
|                | 30 | 91  | 0.93 | 0.03 |

|      |    |     |      |      |
|------|----|-----|------|------|
| GS15 | 0  | 110 | 0.70 | 0.03 |
|      | 10 | 89  | 0.68 | 0.03 |
|      | 20 | 28  | 0.74 | 0.07 |
|      | 60 | 20  | 0.65 | 0.08 |

|               |    |    |      |      |
|---------------|----|----|------|------|
| GFP-golgin-84 | 0  | 78 | 0.05 | 0.05 |
|               | 10 | 33 | 0.04 | 0.08 |
|               | 20 | 52 | 0.10 | 0.15 |
|               | 30 | 48 | 0.40 | 0.11 |

|            |    |    |      |      |
|------------|----|----|------|------|
| CD8a-furin | 0  | 70 | 2.22 | 0.12 |
|            | 10 | 59 | 1.73 | 0.17 |
|            | 20 | 34 | 1.51 | 0.24 |
|            | 30 | 41 | 1.39 | 0.15 |

|              |    |    |      |      |
|--------------|----|----|------|------|
| CD8a-CI-M6PR | 0  | 47 | 2.37 | 0.21 |
|              | 10 | 29 | 1.77 | 0.21 |
|              | 20 | 32 | 1.63 | 0.14 |
|              | 30 | 36 | 1.29 | 0.18 |

Supplementary File 1 (continued)

Fig. S1

A. SBP-GFP-TfR-WT

| Chase time (min) | <i>n</i> | <i>LQ</i> | <i>SEM</i> |
|------------------|----------|-----------|------------|
| 5                | 76       | 0.16      | 0.02       |
| 10               | 95       | 0.51      | 0.03       |
| 15               | 83       | 0.82      | 0.03       |
| 20               | 58       | 0.95      | 0.06       |
| 30               | 37       | 1.10      | 0.08       |
| 40               | 37       | 1.10      | 0.05       |

B. SBP-GFP-Tac #2

|    |     |       |      |
|----|-----|-------|------|
| 5  | 59  | -0.02 | 0.06 |
| 7  | 77  | 0.05  | 0.04 |
| 10 | 78  | 0.23  | 0.05 |
| 15 | 127 | 0.54  | 0.03 |
| 20 | 127 | 0.58  | 0.04 |
| 25 | 132 | 0.70  | 0.04 |
| 30 | 97  | 0.82  | 0.04 |
| 50 | 99  | 0.91  | 0.04 |
| 80 | 96  | 0.89  | 0.05 |

C. SBP-GFP-Tac-TC #2

|     |     |       |      |
|-----|-----|-------|------|
| 5   | 21  | -0.18 | 0.17 |
| 7   | 16  | -0.23 | 0.17 |
| 10  | 36  | 0.19  | 0.07 |
| 15  | 70  | 0.40  | 0.07 |
| 25  | 119 | 0.69  | 0.09 |
| 30  | 127 | 0.77  | 0.03 |
| 40  | 126 | 0.84  | 0.03 |
| 50  | 178 | 0.93  | 0.03 |
| 80  | 112 | 0.99  | 0.02 |
| 110 | 121 | 1.11  | 0.03 |
| 140 | 113 | 1.18  | 0.03 |
| 200 | 182 | 1.19  | 0.02 |

D. SBP-GFP-Ecadherin #2

|    |     |      |      |
|----|-----|------|------|
| 5  | 54  | 0.02 | 0.04 |
| 10 | 55  | 0.31 | 0.04 |
| 20 | 103 | 0.91 | 0.05 |
| 40 | 44  | 1.06 | 0.05 |
| 60 | 36  | 1.03 | 0.17 |
| 70 | 34  | 1.07 | 0.24 |

E. SBP-GFP-Ecadherin #3

|    |    |      |      |
|----|----|------|------|
| 5  | 43 | 0.18 | 0.03 |
| 10 | 43 | 0.52 | 0.06 |
| 20 | 39 | 1.00 | 0.07 |
| 40 | 36 | 1.01 | 0.09 |
| 60 | 11 | 1.14 | 0.23 |
| 70 | 21 | 1.14 | 0.12 |

F. SBP-GFP-CD8a-furin-WT #2

|    |    |      |      |
|----|----|------|------|
| 10 | 48 | 0.04 | 0.02 |
| 20 | 25 | 0.28 | 0.07 |
| 40 | 40 | 1.28 | 0.08 |
| 60 | 49 | 1.34 | 0.07 |

Supplementary File 1 (continued)

F. SBP-GFP-CD8a-furin-WT #2

|     |    |      |      |
|-----|----|------|------|
| 120 | 46 | 1.37 | 0.08 |
| 240 | 31 | 1.47 | 0.13 |
| 360 | 55 | 1.55 | 0.06 |
| 480 | 45 | 1.79 | 0.09 |

G. SBP-GFP-CD8a-furin-WT #3

|     |     |      |      |
|-----|-----|------|------|
| 5   | 89  | 0.02 | 0.02 |
| 10  | 89  | 0.27 | 0.02 |
| 20  | 100 | 0.64 | 0.05 |
| 30  | 116 | 1.32 | 0.04 |
| 40  | 131 | 1.42 | 0.03 |
| 60  | 127 | 1.23 | 0.06 |
| 90  | 90  | 1.51 | 0.05 |
| 120 | 82  | 1.47 | 0.05 |
| 150 | 105 | 1.57 | 0.05 |
| 180 | 108 | 1.56 | 0.05 |

H. SBP-GFP-CD8a-furin-YA #1

|     |     |      |      |
|-----|-----|------|------|
| 10  | 32  | 0.07 | 0.04 |
| 20  | 28  | 1.00 | 0.10 |
| 40  | 59  | 1.51 | 0.07 |
| 60  | 65  | 1.61 | 0.07 |
| 120 | 38  | 1.48 | 0.05 |
| 240 | 100 | 1.54 | 0.04 |
| 360 | 73  | 1.39 | 0.04 |
| 480 | 86  | 1.50 | 0.04 |

I. SBP-GFP-CD8a-furin-YA #3

|     |    |      |      |
|-----|----|------|------|
| 10  | 20 | 0.25 | 0.04 |
| 20  | 15 | 0.63 | 0.12 |
| 40  | 35 | 1.27 | 0.08 |
| 60  | 43 | 1.22 | 0.07 |
| 120 | 46 | 1.28 | 0.08 |
| 240 | 44 | 1.35 | 0.09 |
| 360 | 64 | 1.52 | 0.05 |
| 480 | 48 | 1.67 | 0.07 |

J. SBP-GFP-CD8a-furin-AC #3

|     |    |      |      |
|-----|----|------|------|
| 20  | 35 | 0.80 | 0.07 |
| 40  | 14 | 1.06 | 0.17 |
| 60  | 24 | 1.39 | 0.16 |
| 120 | 11 | 1.62 | 0.14 |
| 150 | 12 | 1.52 | 0.10 |

K. SBP-GFP-CD8a-furin-AC #2

|     |    |      |      |
|-----|----|------|------|
| 20  | 30 | 0.67 | 0.06 |
| 40  | 28 | 1.01 | 0.09 |
| 60  | 25 | 1.21 | 0.08 |
| 120 | 14 | 1.42 | 0.17 |
| 150 | 12 | 1.50 | 0.21 |

L. SBP-GFP-CD8a-furin-Y+AC #2

|    |    |      |      |
|----|----|------|------|
| 10 | 67 | 0.15 | 0.04 |
| 20 | 62 | 0.69 | 0.06 |
| 30 | 27 | 0.68 | 0.07 |
| 40 | 39 | 0.89 | 0.09 |

**Supplementary File 1** (continued)

M. SBP-GFP-CD8a-furin-Y+AC #3

|    |    |      |      |
|----|----|------|------|
| 60 | 28 | 1.00 | 0.08 |
| 70 | 16 | 1.04 | 0.09 |

|    |    |      |      |
|----|----|------|------|
| 10 | 44 | 0.31 | 0.03 |
| 20 | 82 | 0.62 | 0.04 |
| 25 | 89 | 0.95 | 0.04 |
| 30 | 72 | 0.94 | 0.04 |
| 40 | 40 | 0.99 | 0.05 |
| 50 | 49 | 0.97 | 0.05 |

**Supplementary File 1** (continued)

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## Editors

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## Reviewer #1 (Public Review):

Summary:

In the manuscript by Tie et.al., the authors couple the methodology which they have developed to measure LQ (localization quotient) of proteins within the Golgi apparatus along with RUSH based cargo release to quantify the speed of different cargos traveling through Golgi stacks in nocodazole induced Golgi ministacks to differentiate between cisternal progression vs stable compartment model of the Golgi apparatus. The debate between cisternal progression model and stable compartment model has been intense and going on for decades and important to understand the basic way of function/organization of the Golgi apparatus. As per the stable compartment model, cisterna are stable structures and cargo moves along the Golgi apparatus in vesicular carriers. While as per cisternal progression model, Golgi cisterna themselves mature acquiring new identity from the cis face to the trans face and act as transport carriers themselves. In this work, authors provide a missing part regarding intra-Golgi speed for transport of different cargoes as well as the speed of TGN exit and based on the differences in the transport velocities for different cargoes tested favor a stable compartment model. The argument which authors make is that if there is cisternal progression, all the cargoes should have a similar intra-Golgi transport speed which is essentially the rate at which the Golgi cisterna mature. Furthermore, using a combination of BFA and Nocodazole treatments authors show that the compartments remain stable in cells for at least 30-60 minutes after BFA treatment.

Strengths:

The method to accurately measure localization of a protein within the Golgi stack is rigorously tested in the previous publications from the same authors and in combination with pulse chase approaches has been used to quantify transport velocities of cargoes

through the Golgi. This is a novel aspect in this paper and differences in intra-Golgi velocities for different cargoes tested makes a case for a stable compartment model.

Weaknesses:

Experiments are only tested in one cell line (HeLa cells) and predominantly derived from experimental paradigm using RUSH assays where a secretory cargo is released in a wave (not the most physiological condition) and therefore additional approaches would make a more compelling case for the model.

<https://doi.org/10.7554/eLife.98582.1.sa2>

#### **Reviewer #2 (Public Review):**

Summary:

This manuscript describes the use of quantitative imaging approaches, which have been a key element of the labs work over the past years, to address one of the major unresolved discussions in trafficking: intra-Golgi transport. The approach used has been clearly described in the labs previous papers, and is thus clearly described. The authors clearly address the weaknesses in this manuscript and do not overstate the conclusions drawn from the data. The only weakness not addressed is the concept of blocking COPI transport with BFA, which is a strong inhibitor and causes general disruption of the system. This is an interesting element of the paper, which I think could be improved upon by using more specific COPI inhibitors instead, although I understand that this is not necessarily straightforward.

I commend the authors on their clear and precise presentation of this body of work, incorporating mathematical modelling with a fundamental question in cell biology. In all, I think that this is a very robust body of work, that provides a sound conclusion in support of the stable compartment model for the Golgi.

General points:

The manuscript contains a lot of background in its results sections, and the authors may wish to consider rebalancing the text: The section beginning at Line 175 is about 90% background and 10% data. Could some data currently in supplementary be included here to redress this balance, or this part combined with another?

<https://doi.org/10.7554/eLife.98582.1.sa1>

#### **Reviewer #3 (Public Review):**

The manuscript by Tie et al. provides a quantitative assessment of intra-Golgi transport of diverse cargos. Quantitative approaches using fluorescence microscopy of RUSH synchronized cargos, namely GLIM and measurement of Golgi residence time, previously developed by the author's team (publications from 2016 to 2022), are being used here.

Most of the results have been already published by the same team in 2016, 2017, 2020 and 2021. In this manuscript, very few new data have been added. The authors have put together measurements of intra-Golgi transport kinetics and Golgi residence time of many cargos. The quantitative results are supported by a large number of Golgi mini-stacks/cells analyzed. They are discussed with regard to the intra-Golgi transport models being debated in the field, namely the cisternal maturation/progression model and the stable compartments model.

However, over the past decades, the cisternal progression model has been mostly accepted thanks to many experimental data.

The authors show that different cargos have distinct intra-Golgi transport kinetics and that the Golgi residence time of glycosyltransferases is high. From this and the experiment using brefeldinA, the authors suggest that the rim progression model, adapted from the stable compartments model, fits with their experimental data.

#### Strengths:

The major strength of this manuscript is to put together many quantitative results that the authors previously obtained and to discuss them to give food for thought about the intra-Golgi transport mechanism.

The analysis by fluorescence microscopy of intra-Golgi transport is tough and is a tour de force of the authors even if their approach show limitations, which are clearly stated. Their work is remarkable in regards to the numbers of Golgi markers and secretory cargos which have been analyzed.

#### Weaknesses:

As previously mentioned, most of the data provided here were already published and thus accessible for the community. Is there is a need to publish them again?

The authors' discussion about the intra-Golgi transport model is rather simplistic. In the introduction, there is no mention of the most recent models, namely the rapid partitioning and the rim progression models. To my opinion, the tubular connections between cisternae and the diffusion/biochemical properties of cargos are not enough taken into account to interpret the results. Indeed, tubular connections and biochemical properties of the cargos may affect their transit through the Golgi and the kinetics with which they reach the TGN for Golgi exit.

Nocodazole is being used to form Golgi mini-stacks, which are necessary to allow intra-Golgi measurement. The use of nocodazole might affect cellular homeostasis but this is clearly stated by the authors and is acceptable as we need to perturb the system to conduct this analysis. However, the manual selection of the Golgi mini-stack being analyzed raises a major concern. As far as I understood, the authors select the mini-stacks where the cargo and the Golgi reference markers are clearly detectable and separated, which might introduce a bias in the analysis.

The terms 'Golgi residence time' is being used but it corresponds to the residence time in the trans-cisterna only as the cargo has been accumulated in the trans-Golgi thanks to a 20°C block. The kinetics of disappearance of the protein of interest is then monitored after 20°C to 37°C switch.

Another concern also lies in the differences that would be introduced by different expression levels of the cargo on the kinetics of their intra-Golgi transport and of their packaging into post-Golgi carriers.

<https://doi.org/10.7554/eLife.98582.1.sa0>